Maloney's Geomethry

An Axiomatic Formalization of Synthetic Geometry

$$\mathfrak{M} \coloneqq \langle \mathbf{P}, \mathcal{B}, \cong, \mathcal{L}, \mathcal{A} \rangle$$

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Guide to Notation

Objects and Sets

A, B, C, $\ldots$, X, Y, Z
Capital italic: **points**
$\mathbf{P}$ The set of all points
$\mathfrak{A}$, $\mathfrak{B}$, $\mathfrak{C}$, $\ldots \mathfrak{X}$, $\mathfrak{Y}$, $\mathfrak{Z}$ Capital x: geometric figures
$\overline{AB}$ **Line Segment**
$\overleftrightarrow{AB}$ **Line**
$\overrightarrow{AB}$ **Ray**
$\triangle ABC$ **Triangle**
$\angle ABC$ **Angle**
$\overline{AB} \parallel \overline{CD}$ **Parallel**
$\overline{AB} \perp \overline{CD}$ **Perpendicular**

0.1 Primative Relations

$\mathcal{B}(A, B, C)$ Betweeness
$A = B$ Equivalence
$\overline{AB}$, $\overline{CD}$ Congruence

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$\mathcal{L}$ The **Language**
$\mathcal{A}$ The **Axiom Set**

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Specification of primitive symbols and arities defining a formal system.

Definition 1 (Signature of Geometry)**.** *A signature $\mathcal{L}_{geo}$ consists of:*

- *a ternary relation symbol $\mathcal{B}$ (betweenness),*
- *a quaternary relation symbol $\cong$ (congruence).*

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Chapter 6

Primative Notions

For our axiomatic system, we adopt three primative notions, consisting of one object and two relations, which together form a single structure. The first of these is the point, denoted by upper case latin letters, $A, B, C, \ldots$. Each point is a member of the point-set $\mathbf{P}$, which is defined as $\mathbf{P} \coloneqq \{P \mid P \text{ is a point}\}$. This set is equipped with a ternary relation $\mathcal{B}$ (betwenessness) and an quadernary relation $\cong$ (congruence). Together, the domain and these two relations constitute the plane $\mathcal{P}$:

$$\mathcal{P} \coloneqq \langle \mathbf{P}, \mathcal{B}, \cong \rangle.$$

6.1 Plane

The plane is the structure consisting of the domain $\mathbf{P}$ together with the relations $\mathcal{B}$ and $\cong$.

Definition 2 (Plane)**.** *A plane is a structure*

$$\mathcal{P} \coloneqq \langle \mathbf{P}, \mathcal{B}, \cong \rangle.$$

Where $\mathbf{P}$ *is the domain.*

6.2 Points

The elements of the domain $\mathbf{P}$ are termed *points* and denoted using capital italics such as A, B, C, $\ldots$.. At this foundational level, a point is defined not via decomposition into simplier parts, but solely by its membership in this set.

Intuitvely, a point can be understood as a unique position on the plane. This corresponds to the Euclidean charactarize of a point as "that which has no part". A point can be informally regarded as having no magnitude, length, width, or depth, and as being zero-dimensional. However, within the formal system, this physical charactarization is set aside and a point is treated strictly as a primative object with no further definition, thus, the charactarization of these objects carries no weight in the formal deduction.

6.3 Betweenness

Betweenness is a ternary relation on the set of all points, denoted by $\mathcal{B} \subseteq \mathbf{P}^3$. Given points A, B, and C, we write $\mathcal{B}(A, B, C)$ and read it as "B lies between A and C".

Unlike a set $\{A, B, C\}$, where the order of elements is immaterial, the betweeness is defined over the cartesian product $\mathbf{P} \times \mathbf{P} \times \mathbf{P}$. Thus it depends on the order of the arguments. Formally,

$$\mathcal{B} \subseteq \mathbf{P}^3 \coloneqq \{(A, B, C) \mid A, B, C \in \mathbf{P}\}.$$

The expression $\mathcal{B}(A,B,C)$ is simply the assertion that $(A,B,C) \in \mathcal{B}$.

To lighten notation we introduce the square-bracket shorthand:

$$[ABC] \overset{\text{def}}{\iff} \mathcal{B}(A,B,C).$$

begin ai: **Intuitive picture.** In Euclidean geometry a point B is between A and C precisely when the three points lie on a line and B is met while traveling from A to C along that line. Our axioms capture this idea without invoking lines or order directly.

We now impose five axioms that govern the behaviour of $\mathcal{B}$. end ai

We now impose five axioms governing the behavior of $\mathcal{B}$.

Axiom 1 (Identity of Betweeness)**.**

$$\forall A,B \in \mathbf{P}, \mathcal{B}(A,B,A) \implies A = B.$$

This axiom says that if a point appears as the "middle" of a triple that begins and ends with the same point, then the two outer points must coincide. In particular, it forbids the possibity that a distinct point B B lies between A and A; such a configuration would contradict the intuition that betweenness requires a genuine interval. Notice that the converse, $A = B \implies \mathcal{B}(A,A,A)$, is *not* assumed; indeed the next axiom will exclude it.

Axiom 2 (Symmetry of Betweenness)**.**

$$\forall A,B,C \in \mathbf{P}, \mathcal{B}(A,B,C) \implies \mathcal{B}(C,B,A).$$

Betweenness is symmetric with respect to the outer points: if B lies between A and C, then it also lies between C and A. This reflects the fact that the segment AC is the same as the segment CA. Left-to-right implication is used over a bicondinitional due to reduncence, as the right-to-left implication directly follows from this form of the axiom.

Theorem 1. *$\mathcal{B}(A,B,C) \iff \mathcal{B}(C,B,A)$ for all $A,B,C \in \mathbf{P}$.*

Proof. The forward direction is the axiom; the reverse follows by applying the axiom with A and C exchanged. □

Axiom 3 (Order)**.** $\forall A,B,X \in \mathbf{P},\ \mathcal{B}(A,B,X) \implies \big(\neg\mathcal{B}(B,A,X) \wedge \neg\mathcal{B}(A,X,B)\big)$.

If B is between A and X, then A cannot be between B and X, nor can X be between A and B. In other words, a point can serve as a "middle" in only one way: the triple (A,B,X) determines a unique order along the line. Together with the Identity axiom, this implies that $\mathcal{B}(A,A,A)$ is impossible (otherwise, taking $A = B = X$, we would have both $\mathcal{B}(A,A,A)$ and its negation). Thus the relation is *non-reflexive* in the strict sense: no point lies between itself and itself.

Axiom 4 (Density (inner point))**.** $\forall A,C \in \mathbf{P}, A \neq C \implies \exists B \in \mathbf{P} \colon \mathcal{B}(A,B,C)$.

Whenever two points are distinct, there exists a point strictly between them. This captures the idea that segments are dense—between any two points one can always find a third.

Axiom 5 (Extensibility (outer point))**.** $\forall A,B \in \mathbf{P},\ A \neq B \implies \exists C \in \mathbf{P} \colon\ \mathcal{B}(A,B,C)$.

Given two distinct points, there is always a point beyond the second, so that the second lies between the first and the new point. This guarantees that lines extend indefinitely.

These five axioms are mutually independent: no one can be derived from the others (as can be shown by standard model-theoretic arguments). Together with the congruence axioms to be introduced later, they will form the basis of our geometric theory.

Theorem 2 (Irreflexivity of Betweeness)**.** *No point lies between it and itself:*

$$\forall A \in \mathbf{P},\ \neg\mathcal{B}(A, A, A).$$

Proof. Assume $\mathcal{B}(A, A, A)$. By the order axiom, this implies $\neg\mathcal{B}(A, A, A)$ and $\neg\mathcal{B}(A, A, A)$, which contridicts our assumption. Hence, $\neg\mathcal{B}(A, A, A)$. □

Theorem 3 (rict BetweeDistinct Endpoints)**.** *Betweenness forces disctint outer points:*

$$\mathcal{B}(A, B, C) \implies A \neq C.$$

Proof. Suppose $\mathcal{B}(A, B, C)$ and $A = C$. Then, we have $]mathcal B(A, B, A)$, which by the identity of betweenness implies $A = B$. Substituting back gives $\mathcal{B}(A, A, A)$, which contradicts the irreflexivity of betweenness. Therefore $A \neq C$. □

Theorem 4 (St ness)**.** *a*

Proof. □

Theorem 5 (Contrapositive Ordering)**.** *w*

Proof. □

Theorem 6 (Extension of Symmetry)**.** *w*

Proof. □

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2.6 Density and extensibility give infinite sequences These are more observations than theorems:

From Density you can, by repeated application, produce infinitely many points between any two distinct points.

From Extensibility you can extend a segment outward indefinitely.

You can formulate them as remarks or simple lemmas.

6.4 Congruence

Congruence is a quaternary relation on the set of all points denoted by $\cong \subseteq \mathbf{P}^4$. Rather than mapping individual points to one another, this relation establishes a geometric equivalence between two pairs of points. We write $\cong (A, B, C, D)$, to assert that the segment from point A to point B is congruent to the segment from point C to point D. Formally:

$$\cong (A, B, C, D) \iff (A, B, C, D) \in \cong .$$

To reflect this conceptually, we bundle the point pairs (A, B) and (C, D) into the typographical units AB and CD and introduce the infix symbol $\cong$:

$$AB \cong CD \overset{\text{def}}{\iff} \cong (A, B, C, D)$$

If the relation holds for the pairs (A, B) and (C, D), we write $(A, B) \cong (C, D)$, which is interpreted as the "distance" from A to B being equal to the distance from C to D.

We now impose three axioms governing the behavior of $\cong$.

Axiom 6 (Reversal of Congruence)**.**

$$\forall A, B \in \mathbf{P},\ AB \cong BA.$$

Axiom 7 (Reflexivity of Congruence)**.**

$$\forall A, B \in \mathbf{P},\ AB \cong AB.$$

Axiom 8 (Transitivity of Congruence)**.**

$$\forall A, B, C, D, E, F \in \mathbf{P},\ (AB \cong CD \wedge AB \cong EF) \implies CD \cong EF.$$

Theorem 7 (Symmetry of Congruence)**.** *If $AB \cong CD$, then $CD \cong AB$.*

Proof. We are given $AB \cong CD$. From the reflexivity of congruence theorem we also have $AB \cong AB$. Applying these two congruences to the transivity axiom gives

$$AB \cong CD \wedge AB \cong AB,$$

implying $CD \cong AB$. □

Theorem 8 (Left-Reversal of Congruence)**.** *If $AB \cong CD$, then $BA \cong CD$.*

Proof. We are given $AB \cong CD$. From the reversal of congruence, we also have $AB \cong BA$. Applying these to the transitivity of congruence yields $(AB \cong BA \vee AB \cong CD) \implies BA \cong CD$, thus proving the result. □

Theorem 9 (Right-Reversal of Congruence)**.** *If $AB \cong CD$, then $AB \cong DC$.*

Proof. We are given $AB \cong CD$. From the reversal of congruence, we have $CD \cong DC$, and by the symmetry of congruence we have $CD \cong AB$. Applying these to the transitivity of congruence gives the antecedent

$$CD \cong AB \wedge CD \cong DC,$$

which implies $AB \cong DC$. □

Theorem 10 (Full-Reversal of Congruence)**.** *If $AB \cong CD$, then $BA \cong DC$.*

Proof. We are given $AB \cong CD$. We first apply left-reversal to the expression to give $BA \cong CD$. This is repeating but with right-reversal to obtain $BA \cong DC$. □

Proof. We are given $AB \cong CD$. From the reversal of congruence, we know that $AB \cong BA$. Similarily, by right-reversal we have $CD \cong DC$. Applying these to the transitivity of congruence gives

$$AB \cong BA \wedge AB \cong DC.$$

This implies $BA \cong DC$. □

Theorem 11 (Full-Transitivity of Congruence)**.** *If $AB \cong CD$ and $CD \cong EF$, then $AB \cong EF$*

Proof. We first apply symmetry to the first premise giving $CD \cong AB$. Then, we apply this result alongwith the second premise to the partial-transitivity of congruence axiom, giving the antecedent

$$CD \cong AB \wedge CD \cong EF.$$

This implies $AB \cong EF$, thus proving the result. □

With this result proven, we have demostrated that $\cong$ is an equivalence relation on ordered pairs of points, as it is reflexive and symmetric by axiom, and it follows from those axioms that it is fully transitive.

Because $\cong$ is an equivalence relation, it partitions the set of all ordered point pairs into disjoint equivalence classes. Each class consists of all segments that are congruent to one another, which in geometric terms corresponds to the set of all segments having the same length.

Chapter 7

Partition of the Plane

7.1 Plane Separation Axiom

The previous axioms of betweenness govern the linear ordering of points and are therefore limited to a single dimension. To extend this to two dimensions, we introduce the Plane Separation Axiom. This asserts that a line acts as a parition splitting the plane into two discint, non-overlapping regions, which is then used to describe a nototion of points membership to one side of a given line through its logical equivalence to the negation of the previously defined notion of segment intersection.

Axiom 9 (Plane Separation)**.** *For every line $l \in \mathbf{L}$, there exist two sets $H_{l,A}, H_{l,B} \subset \mathbf{P} \setminus l$ such that:*

1. $H_{l,A} \cup H_{l,B} = \mathbf{P} \setminus l \wedge H_{l,A} \cap H_{l,B} = \emptyset$.
2. $\forall A, B \in \mathbf{P} \setminus l \colon (\exists X \in l, \mathcal{B}(A, X, B) \iff \{A, B\} \not\subset H_{l1} \wedge \{A, B\} \not\subset H_{l2})$.

explain the meaning-of-axioms-here

7.2 Same-Side Relation

We formalize the concept of 'being on the same side' through the Same-Side Relation ($\mathcal{S}$). It is defined by the absence of a seperation/parition of the plane: two points are related if and only if unique segment connecting them does not contain any point of the boundary line l. This definition ensures the relation is based upon the primative notion of betweeness. Specifically, we define $\mathcal{S}$ as a subset of the cartesian product $\mathcal{S} \subseteq \mathbf{P} \times \mathbf{P} \times \mathbf{L}$.

Definition 3 (Same-Side Relation)**.** *Given a line $l \in \mathbf{L}$, any two points $A, B \in \mathbf{P} \setminus l$ are related by $\mathcal{S}$ if there does not exist a point $X \in l$ such that $\mathcal{B}(A, X, B)$. That is,*

$$\mathcal{S}(A, B, l) \iff \nexists X \in l \colon \mathcal{B}(A, X, B).$$

7.3 Half-Planes

By grouping all points related to a reference point A by $\mathcal{S}$, we construct the geometrical object known as the half-plane ($H_{l,A}$). This transition allows us to treat regions of the plane as sets that can be intersected to form more complex figures, such as angles and triangles.

Definition 4 (Half-Plane)**.** *Let $l \in \mathbf{L}$ be a line and let $A \in \mathbf{P}$ be a reference point such that $A \notin l$. The half-plane detirmined by l and A, denoted $H_{l,A}$, is the subset of the point-set consistening of all points on the same side of l as A. That is,*

$$H_{l,A} \coloneqq \{X \in \mathbf{P} \setminus l \mid \mathcal{S}(A, X, l)\}.$$

This geometric object allows us to move beyond comparisons between single points. A point X being on the same same as point Y is the same as X being an element of the half-plane $H_{l,y}$. This will show incredibily useful in defining derived geometric figures.

7.4 Theorems

Theorem 12 (Same-Side Reflexivity)**.** *For any point $A \in \mathbf{P}$ that is not a member of any given line $l \in \mathbf{L}$, the following must hold:*

$$\forall l \in \mathbf{L}, \forall A \in \mathbf{P} \colon \mathcal{S}(A, A, l).$$

Proof. Let there be a line $l \in \mathbf{L}$ and a point $A \in \mathbf{P} \setminus l$. We proceed by contradiction. Assuming the same-side relation $\mathcal{S}(A, A, l)$ is false, it follows that $\nexists X \in l \colon \mathcal{B}(A, X, A)$ is false by definition, and that therefore there does exist an $X \in l$ satisfing these conditions. As a consequence, by the Idenity/Reflexivity of Betweenness $A = X$.

However, we have defined that $A \notin l$. Since our assumption leads to a contradiction, it must be that no such point X exists. Thus, $\mathcal{S}(A, A, l)$ is true. □

Theorem 13 (Same-Side Symmetry)**.**

$$\mathcal{S}(A, B, l) \iff \mathcal{S}(B, A, l).$$

Proof. Firstly, we note that $\mathcal{S}(A, B, l) \iff \nexists X \in l \colon \mathcal{B}(A, X, B)$ by definition. By the Symmetry of Betweeness, the right-hand side of this is equivalent to $\nexists X \in l \colon \mathcal{B}(B, X, A)$, thus showing

$$X \in l \colon \neg\mathcal{B}(A, X, B) \iff X \in l \colon \neg\mathcal{B}(B, X, A).$$

The hand-side is by definition equivalent to $\mathcal{S}(B, A, l)$. Thus, by the transitivity of the biconditional, we conclude $\mathcal{S}(A, B, l) \iff \mathcal{S}(B, A, l)$. □

Theorem 14 (Same-Side Transitivity)**.** *Given any line $l \in \mathbf{L}$ and points $A, B, C \in \mathbf{P} \setminus l$, the relation $\mathcal{S}$ satisfies:*

$$\mathcal{S}(A, B, l) \wedge \mathcal{S}(B, C, l) \implies \mathcal{S}(A, C, l).$$

Proof. By the Plane Seperation Axiom, we know there must exist two distint sets H_{l1} and H_{l2} whose union is $\mathbf{P} \setminus l$. Since A, B, and C are not on l, each must belong to exactly one of these two sets.

Assume $\mathcal{S}(A, B, l)$. By definition, they are contained within the same set. Without loss of generality, let $A, B \in H_{l1}$.

Assume $\mathcal{S}(B, C, l)$. Similarily, this implies they are in the same set. We previously established that $B \in H_{l1}$, implying that $C \in H_{l1}$ by the plane seperation axiom ($H_{l1} \cap H_{l2} = \emptyset$).

We now have $A \in H_{l1}$ and $C \in H_{l1}$. By the inverse of PSA2, there does not exist a point $X \in l$ such that $\mathcal{B}(A, X, C)$. Thus, by definition, A and C must be related by the same-side relation. That is, $\mathcal{S}(A, C, l)$. □

randomideas:

Chapter 8

Angles

As opposed to being a measure, we define an angle as angle as a geometric figure, ie: a set of points. Specifically, an angle is the union of two rays that share a common vertex.

Definition 5 (Angle)**.** *An* ***angle*** *is the union of two rays* $\overrightarrow{AO}$ *and* $\overrightarrow{OB}$ *that share a common vertex* O*, denoted as follows*

$$\angle AOB \coloneqq \overrightarrow{AO} \cup \overrightarrow{OB}.$$

Using the half-planes defined previously, we have the tools neccessary to describe the interior and exterior of an angle.

Definition 6 (Interior of an Angle)**.** *For any three non-collinear points* $A, B, C \in \mathbf{P}$*, the* ***interior*** *of* $\angle ABC$ *is the set:*

$$\mathit{Int}(\angle ABC) \coloneqq \{P \in \mathbf{P} \mid \mathcal{S}(P, A, \overleftrightarrow{BC}) \wedge \mathcal{S}(P, C, \overleftrightarrow{BA})\}.$$

Equivalently, this is the intersection of two half-planes:

$$\mathit{Int}(\angle ABC) \coloneqq H_{\overleftrightarrow{BC},A} \cap H_{\overleftrightarrow{BA},C}.$$

Definition 7 (Interior of an Angle)**.** *For any three non-collinear points* $A, B, C \in \mathbf{P}$*, the* ***interior*** *of* $\angle ABC$ *is the set:*

$$\mathit{Int}(\angle ABC) \coloneqq \{P \in \mathbf{P} \mid \mathcal{S}(P, A, \overleftrightarrow{BC}) \wedge \mathcal{S}(P, C, \overleftrightarrow{BA})\}.$$

Equivalently, this is the intersection of two half-planes:

$$\mathit{Int}(\angle ABC) \coloneqq H_{\overleftrightarrow{BC},A} \cap H_{\overleftrightarrow{BA},C}.$$

Chapter 9

Linear Notions

9.1 Defintion of Linear Notions

A segment is a subset of $\mathbf{P}$ defined as the union of two discint points and all points between them. Symbolically, that is

Definition 8 (Segment)**.**

$$\forall A, B \in \mathbf{P}, A \neq B \implies AB \coloneqq \{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(A, X, B)\}.$$

A ray pointing from point A to a discint point B is a subset of $\mathbf{P}$ denoted as $\overrightarrow{AB}$. It is defined as the union of A, B, all points between A and B, and all points such that B is between A and that point, that is

Definition 9 (Ray)**.**

$$\forall A, B \in \mathbf{P}, A \neq B \implies \overrightarrow{AB} \coloneqq \{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(A, X, B) \vee \mathcal{B}(A, B, X)\}.$$

The line $\overleftrightarrow{AB}$ is the subset of $\mathbf{P}$ consisting of the points A, B, all points between A and B, all points where A is between it and B, and all points where B is between it and A. That is,

Definition 10 (Line)**.**

$$\forall A, B \in \mathbf{P}, A \neq B \implies \overleftrightarrow{AB} \coloneqq \{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(X, A, B) \vee \mathcal{B}(A, X, B) \vee \mathcal{B}(A, B, X)\}.$$

9.2 Properties of Linear Notions

Theorem 15 (Distinctness of Betweenness)**.** *If $\mathcal{B}(A, B, X)$ holds, then A, B, and X are three distinct points:*

$$\forall A, B, X \in \mathbf{P}, \mathcal{B}(A, B, X) \implies (A \neq B \wedge B \neq X \wedge A \neq X).$$

Proof. We proceed by contradiction. Suppose $\mathcal{B}(A, B, X)$ holds.

First, assume $A = X$. Substituting A for X in our hypothesis, we obtain $\mathcal{B}(A, B, A)$, which implies $A = B$. If $A = B$, then the relation $\mathcal{B}(A, B, X)$ is equivalent to $\mathcal{B}(B, A, X)$. However, by the Order of Betweeness, $\mathcal{B}(A, B, X) \implies \neg\mathcal{B}(B, A, X)$. Here, we reach a contradiction and conclude our hypothesis must be false and thus $A \neq X$.

Next, we test the assumption $A = B$. Under this assumption, we substite our hypothesis to obtain $\mathcal{B}(B, A, X)$. Similarly, the Order of Betweeness shows these cannot be simutanously true, and thus our assumption must be false, meaning $A \neq B$.

Finally, by the Transitivity of Equality, $B \neq X$. Thus proving the result. $\square$

Theorem 16 (Segment Symmetry)**.** *For any two points A, B, the segment AB is equal to the segment BA.*

$$\forall A, B \in \mathbf{P}, AB = BA.$$

Proof. By the definition of a line segment, we have:

$$AB = \{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(A, X, B)\}$$

and

$$BA = \{B, A\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(B, X, A)\}.$$

First, by the commutative property of sets, the endpoints are identical: $\{A, B\} = \{B, A\}$.

Second, we examine the sets of interior points. By the **Symmetry of Betweenness**, the relation $\mathcal{B}(A, X, B)$ holds if and only if $\mathcal{B}(B, X, A)$ holds. Formally, this implies:

$$\{X \in \mathbf{P} \mid \mathcal{B}(A, X, B)\} = \{X \in \mathbf{P} \mid \mathcal{B}(B, X, A)\}.$$

Since AB and BA are defined as the union of identical sets, they contain the same points. By the principle of extensionality in set theory, $AB = BA$. □

Theorem 17 (Line Symmetry)**.** *For any two distinct points A and B, the line $\overleftrightarrow{AB}$ is equal to the segment $\overleftrightarrow{BA}$:*

$$\forall A, B \in \mathbf{P}, \overleftrightarrow{AB} = \overleftrightarrow{BA}.$$

Proof. By the definition of a line, we have:

$$\overleftrightarrow{AB} = \{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(X, A, B) \vee \mathcal{B}(A, X, B) \vee \mathcal{B}(A, B, X)\}$$

and

$$\overleftrightarrow{BA} = \{B, A\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(X, B, A) \vee \mathcal{B}(B, X, A) \vee \mathcal{B}(B, A, X)\}.$$

First, by the commutative property of sets, the endpoints are identical: $\{A, B\} = \{B, A\}$.

Second, we demonstrate that the three cases of betweeness in $\overleftrightarrow{AB}$ are logically equivalent to those of $\overleftrightarrow{BA}$ using the by the **Symmetry of Betweenness**:

- $\{X \in \mathbf{P} \mid \mathcal{B}(X, A, B)\} = \{X \in \mathbf{P} \mid \mathcal{B}(B, A, X)\}$
- $\{X \in \mathbf{P} \mid \mathcal{B}(A, X, B)\} = \{X \in \mathbf{P} \mid \mathcal{B}(B, X, A)\}$
- $\{X \in \mathbf{P} \mid \mathcal{B}(A, B, X)\} = \{X \in \mathbf{P} \mid \mathcal{B}(X, B, A)\}$

Since each condition for membership in $\overleftrightarrow{AB}$ is logically equivalent to a condition for membership in $\overleftrightarrow{BA}$, the sets contain the same points. Thus, by the principle of extensionality, $\overleftrightarrow{AB} = \overleftrightarrow{BA}$ □

Theorem 18 (Ray Asymmetry)**.** *For any two distinct points $A, B \in \mathbf{P}$, the ray $\overrightarrow{AB}$ is not equal to the ray $\overleftarrow{BA}$:*

$$\forall A, B \in \mathbf{P}, A \neq B \implies \overrightarrow{AB} \neq \overrightarrow{BA}.$$

Proof. We proceed by by contradiction. Assume that the rays for some distinct points $A, B \in \mathbf{P}$, $\overrightarrow{AB} = \overrightarrow{BA}$.

By the definition of a ray, we have:

$$\overrightarrow{AB} = \{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(A, X, B) \vee \mathcal{B}(A, B, X)\},$$

and

$$\overrightarrow{BA} = \{B, A\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(B, X, A) \vee \mathcal{B}(B, A, X)\}.$$

By the **Outer Point Axiom**, there must exist a point Y such that $\mathcal{B}(A, B, Y)$. By the defintion of $\overrightarrow{AB}$, it follows that $Y \in \overrightarrow{AB}$. Under our initial assumption that the rays, Y must be an element of $\overrightarrow{BA}$.

By the Distinctness of Betweeness, the relation $\mathcal{B}(A, B, Y)$ implies that $A \neq Y$ and $B \neq Y$. Therefore, $Y \notin \{A, B\}$.

Since $Y \in \overrightarrow{BA}$ but is not an endpoint, it must satisfy $\mathcal{B}(B, Y, A)$ or $\mathcal{B}(B, A, Y)$.

However, because we know that $\mathcal{B}(A, B, Y)$ is true, by Order of Betweeness we have $\mathcal{B}(A, B, Y) \implies \neg\mathcal{B}(B, A, Y)$, and $\mathcal{B}(A, B, Y) \implies \neg\mathcal{B}(A, Y, B)$, which by the Symmetry of Betweeness is logically equivalent to $\mathcal{B}(A, B, Y) \implies \neg\mathcal{B}(B, Y, A)$.

Thus, both conditions for Y to be an element of $\overrightarrow{BA}$ are false. Therefore, the rays $\overrightarrow{AB}$ and $\overrightarrow{BA}$ are not equal, thus proving the desired result. □

Theorem 19 (Union of Opposite Rays)**.** *For any two points $A, B \in \mathbf{P}$,*

$$\overleftarrow{AB} \cup \overrightarrow{AB} = \overleftrightarrow{AB}.$$

Proof. By the definition of a ray, we have:

$$\overleftarrow{AB} = \{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(X, A, B) \vee \mathcal{B}(A, X, B)\},$$

and

$$\overrightarrow{AB} = \{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(A, X, B) \vee \mathcal{B}(A, B, X)\}.$$

The union of $\overleftarrow{AB}$ and $\overrightarrow{AB}$ therefore consists of the set, $\{B, A\}$, and all points X such that $\mathcal{B}(X, A, B)$ or $\mathcal{B}(A, X, B)$, or $\mathcal{B}(A, B, X)$.

Note that this is is the same as the line $\overleftrightarrow{AB}$, as it is defined as the set

$$\overleftrightarrow{AB} = \{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(X, A, B) \vee \mathcal{B}(A, X, B) \vee \mathcal{B}(A, B, X)\}.$$

This set is equal to the union described above. Thus, the union of the rays $\overleftarrow{AB}$ and $\overrightarrow{AB}$ is equal to the line $\overleftrightarrow{AB}$. □

Theorem 20 (Intersection of Opposite Rays)**.** *For any two points $A, B \in \mathbf{P}$,*

$$\overleftarrow{AB} \cap \overrightarrow{AB} = AB.$$

Proof. By the definition of a ray, we have:

$$\overleftarrow{AB} = \{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(X, A, B) \vee \mathcal{B}(A, X, B)\},$$

and

$$\overrightarrow{AB} = \{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(A, X, B) \vee \mathcal{B}(A, B, X)\}.$$

The intersection of $\overleftarrow{AB}$ and $\overrightarrow{AB}$ therefore consists of the set, $\{B, A\}$ and all points X betweem A and B, that is $\mathcal{B}(A, X, B)$.

This intersection consists of the same points contained in the segment AB, as

$$AB = \{A, B\} \cup \{X \in \mathbf{P} \mid \mathcal{B}(A, X, B)\}.$$

Thus, the union of the rays $\overleftarrow{AB}$ and $\overrightarrow{AB}$ is equal to the segment AB. □

9.3 Parallelism

Two lines are parallel if they have no point in common. In our set-theoretic foundations, this is to mean that the intersection of the lines is the empty set. Because we define parallelism as a binary relation on lines, we first must have a set of all lines to work with. Thus, we let $\mathbf{P}$ be the domain called the set of all points and define $\mathbf{L}$ to be the collection of all subsets of $\mathbf{P}$ that consist of the line $\overleftarrow{AB}$ for a any given two points $A, B \in \mathbf{P}$. Formally, that is:

$$\mathbf{L} \coloneqq \{l \subseteq \mathbf{P} \mid \exists A, B \in \mathbf{P}(A \neq B \land l = \overleftrightarrow{AB})\}.$$

Definition 11 (Parallel)**.** *Given any lines $l, m \in \mathbf{L}$, we say l is parallel to m and write $l \parallel m$ if and only if they are the same same set of points or their set intersection is empty.*

$$\forall l, m \in \mathbf{L},\, l \parallel m \iff (l = m) \lor (l \cap m = \emptyset).$$

add more here

Chapter 10

Triangles

10.1 Defintion

define as the union of three line segments. the three segments must be unique.

define inside and outside of the triangle based on if a point x is on the same side of each comprising segment or not

Chapter 11

Quadrilaterals

motivation

11.1 Definition

A **quadrilateral is defined as the union of four unique noncollinear segments as follows:**

$$\square ABCD := AB \cup BC \cup CD \cup DA.$$

Each line segment is called a side of the quadrilateral, and each point A, B, C, D **is a vertex**

Theorem 21. *idk*

i hate this project lmao ts straight retarded. idk what to do with it ever.

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